

Expt 7 Spectral Estimation

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

PROGRAM

```
clc; clear; close all
```

```
% Given specifications
```

```
Rp = 1;    % Passband Ripple (dB)
```

```
Rs = 20; % Stopband Attenuation (dB)
```

```
Fp = 1000; % Passband Frequency (Hz)
```

```
Fst = 3000; % Stopband Frequency (Hz)
```

```
Fs = 8000; % Sampling Frequency (Hz)
```

```
% Normalized frequencies
```

```
Wp = Fp / (Fs/2);
```

```
Ws = Fst / (Fs/2);
```

```
% Determine minimum filter order and cutoff frequency
```

```
[N, Wn] = buttord(Wp, Ws, Rp, Rs);
```

```
% Design Butterworth Low-Pass Filter
```

```
[b, a] = butter(N, Wn, 'low');
```

```
% Frequency response
figure;
freqz(b, a);
title('Frequency Response of Butterworth LPF');
```

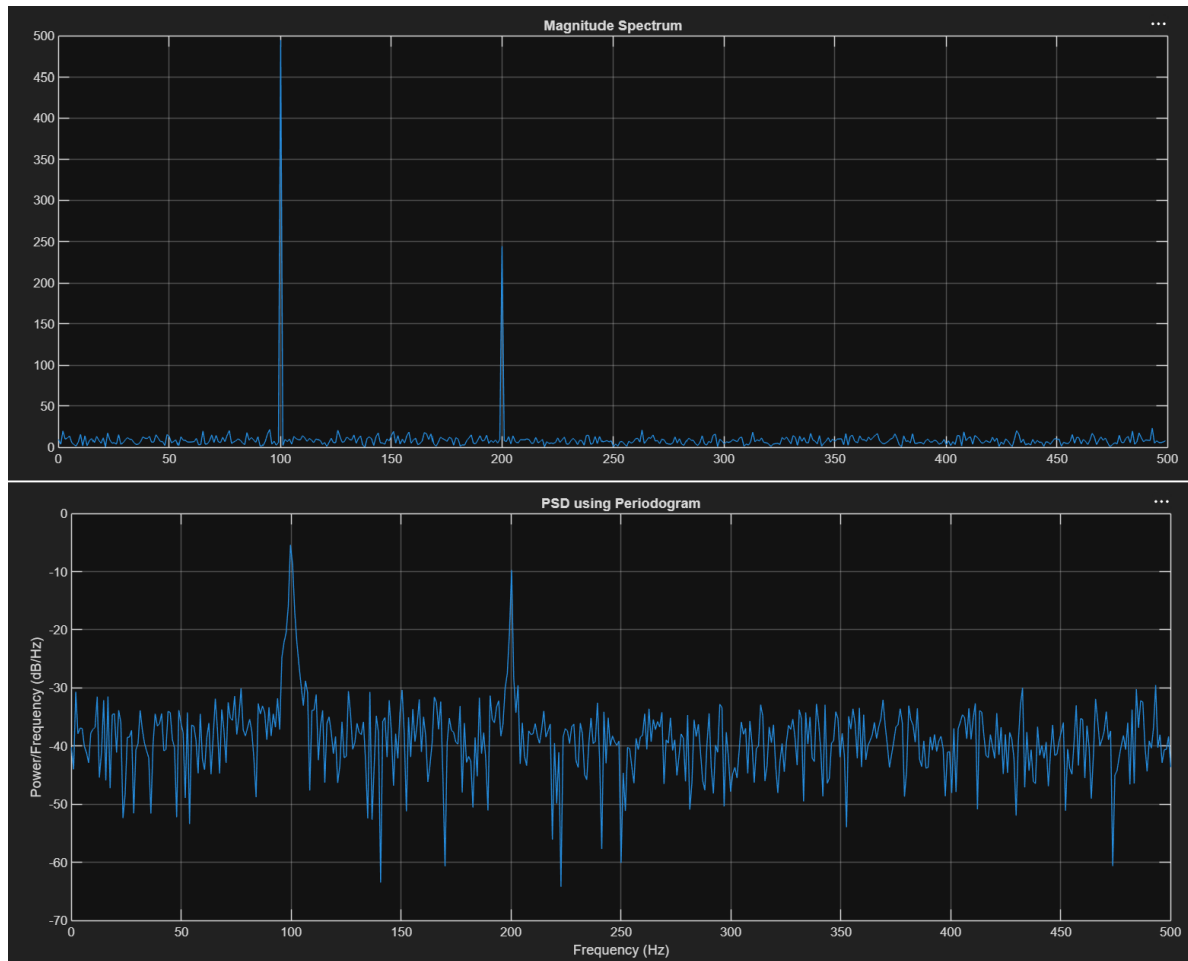
```
% Impulse response
figure;
[h, n] = impz(b, a, 50);
stem(n, h);
title('Impulse Response');
```

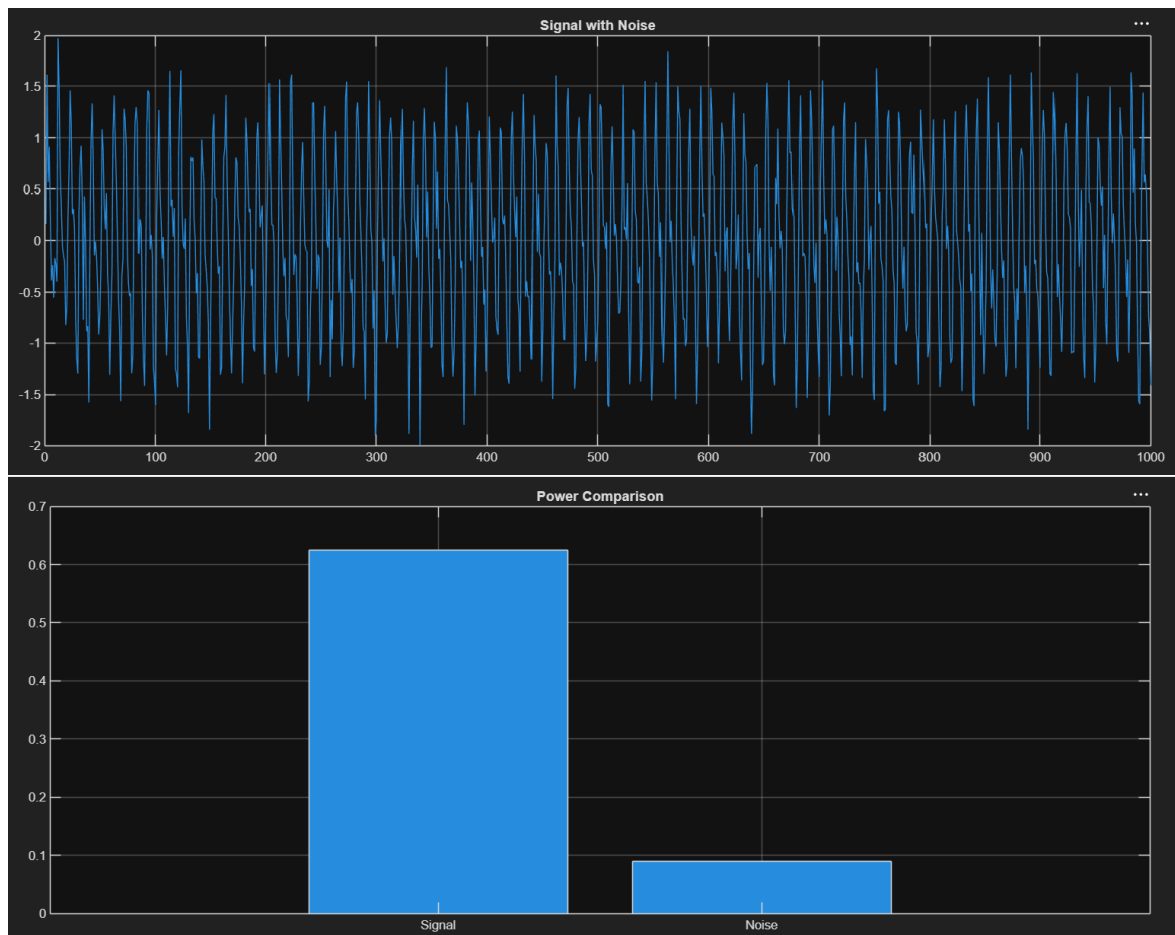
```
% Step response
figure;
stepz(b, a);
title('Step Response');
```

```
% Pole-zero plot
figure;
zplane(b, a);
title('Pole-Zero Plot');
```

```
% Display results
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

OUTPUT





7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, Hanning Windows in MATLAB

PROGRAM

```
clc; clear; close all;
```

```
% Signal
```

```
fs = 1000; N = 256; t = (0:N-1)/fs;
```

```
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
```

```
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);
```

```
% Windows
```

```

w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};
% Processing
for i = 1:3
    xw = x(:).*w{i};
    X = fft(xw,NFFT);
    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i}/max(P{i});
end
%% Individual Spectrums
figure;
for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}),'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window']);
    xlim([0 250]); grid on;
end
%% Comparison
figure;
plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','Li
neWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
%% Window Shapes
figure;
for i = 1:3
    subplot(3,1,i); plot(w{i},'LineWidth',1.5);
    title([names{i} ' Window']); grid on;
end

```

```

%% Peak Detection
for i = 1:3
    [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
    fprintf('\n%s Window Frequencies:\n',names{i});
    disp(loc{i});
end

%% Peak Marking (Hamming)
figure;
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;

%% Main Peaks
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
    [~,idx] = max(P{i});
    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end

```

OUTPUT

Main Spectral Peaks:
 Rectangular Window Peak = 100.10 Hz
 Hamming Window Peak = 100.10 Hz
 Hanning Window Peak = 100.10 Hz

